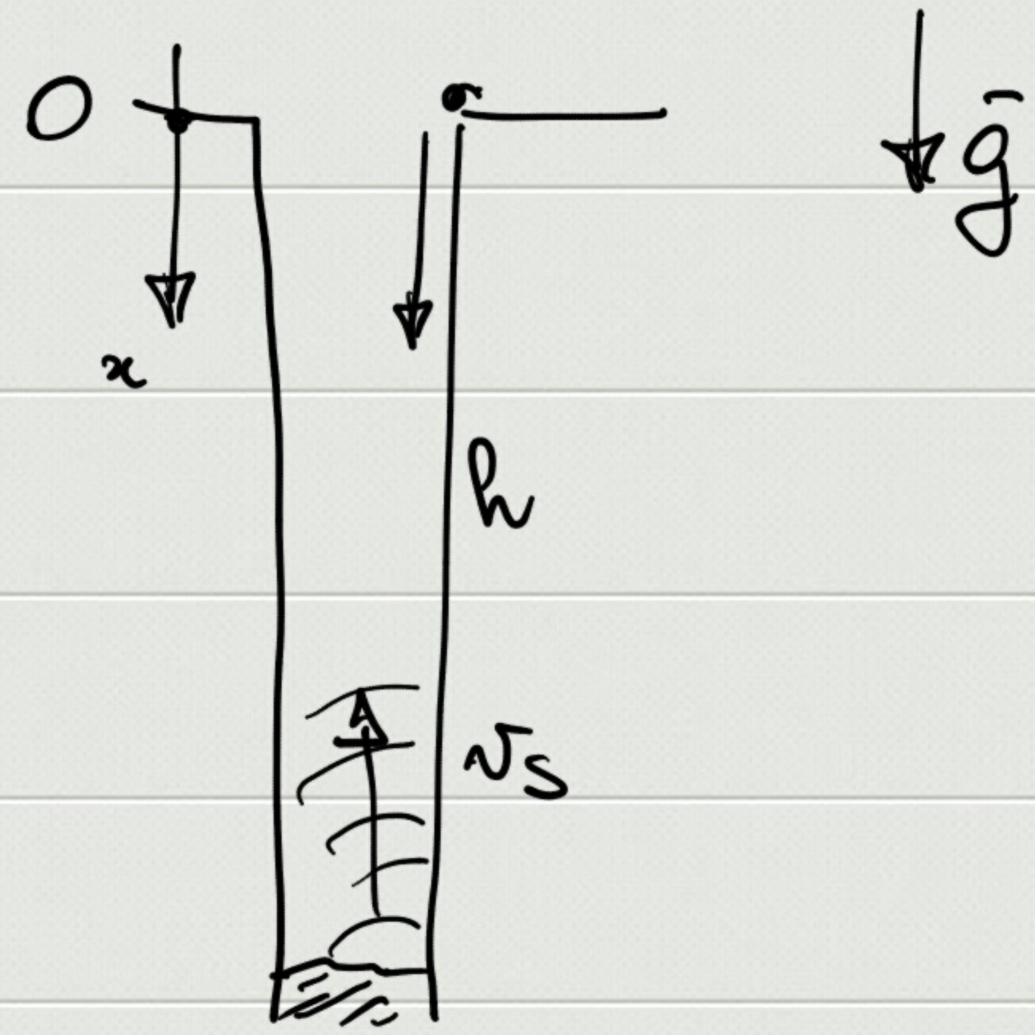
$T = 3.5\ s$

$v_s = 330\ m/s$

$h = ?$



$$T = t_1 + t_2$$

$$h = \frac{1}{2} g t_1^2 \Rightarrow t_1 = \sqrt{\frac{2h}{g}}$$

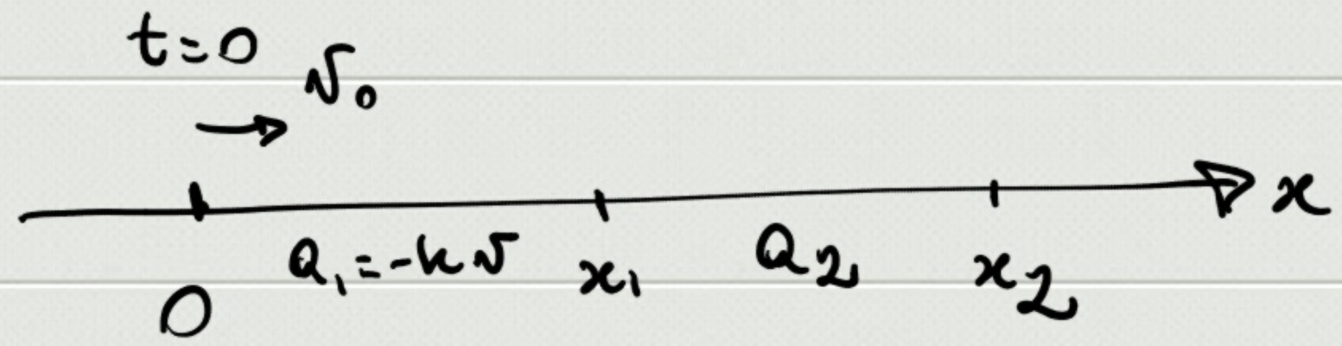
$$h = v_s t_2 \Rightarrow t_2 = \frac{h}{v_s}$$

$$T = \sqrt{\frac{2h}{g}} + \frac{h}{v_s} \Rightarrow T - \frac{h}{v_s} = \sqrt{\frac{2h}{g}}$$

$$T^2 + \frac{h^2}{v_s^2} - \frac{2hT}{v_s} = \frac{2h}{g}$$

$$h^2 - 2h\left(v_s T + \frac{v_s^2}{g}\right) + T^2 v_s^2 = 0$$

$$h = \left(v_s T + \frac{v_s^2}{g}\right) \pm \sqrt{\left(v_s T + \frac{v_s^2}{g}\right)^2 - T^2 v_s^2} = \begin{cases} \boxed{55\ m} \\ \cancel{24\ km} \end{cases}$$



$t_0 = 0 \quad x_0 = 0 \quad v(t_0) = v_0 = 14 \text{ m/s}$

$t > 0 \quad a_1 = -kv \quad k = 2.4 \text{ s}^{-1}$

$[0, x_1] \quad x_1 = 4 \text{ m}$

$[x_1, x_2] \quad x_2 = 8 \text{ m} \quad a_2 = \text{cost} > 0$

$x = x_2 \quad v(x_2) = v_0$

$a_2 = ?$

- $v(x_1) = v_0 + a_1 t_1$ NO $a_1 \neq \text{cost}$
- $v^2(x_1) = v_0^2 + 2 a_1 x_1$ NO
- $v^2(x_1) = v_0^2 + 2 \int_0^{x_1} a_1(x)\, dx$
- $a = v \dfrac{dv}{dx} = -kv$ *

$$\left.\begin{aligned} a = \frac{dv}{dt} &= \frac{dv}{dx}\frac{dx}{dt} = \cancel{v}\frac{dv}{dx} \\ &= -k\cancel{v} \end{aligned}\right\} \quad \boxed{dv = -k\,dx}$$

$$\int_{v_0}^{v_1} dv = -k\int_0^{x_1} dx \Rightarrow \boxed{\overset{v(x_1)}{v_1} = v_0 - k x_1}$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$v^2(x_2) = v_0^2 = v_1^2 + 2a_2(x_2 - x_1)$$

$$\Rightarrow a_2 = \frac{v_0^2 - v_1^2}{2(x_2 - x_1)} =$$

$$= \frac{\cancel{v_0^2} - \cancel{v_0^2} + 2kv_0x_1 - k^2x_1^2}{2(x_2 - x_1)} =$$

$$= \frac{kx_1(2v_0 - kx_1)}{2(x_2 - x_1)} = 22.1 \ m/s^2$$